

Food Supplements for Joint Pain and Arthritis: What Really W

An evidence-based guide to the best joint supplements available in the UK — from glucosamine to omega-3.

By **Living With Arthritis Editorial Team** · Reviewed by **UK-registered health professional** (Health Professional) · 2026-05-27 · livingwitharthritis.org.uk

Walk into any UK pharmacy or health food shop and you will find an entire aisle devoted to joint supplements. From glucosamine and chondroitin to turmeric capsules and collagen powders, the market is enormous — and confusing. With millions of UK adults living with arthritis-related joint pain, understanding which food supplements for joint pain have genuine evidence behind them has never been more important.



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This guide cuts through the marketing noise and focuses on the supplements with the strongest scientific evidence, their recommended dosages, what to look for on labels, and important safety information. We also distinguish between supplements likely to help with osteoarthritis versus those better suited to inflammatory arthritis types.

Who Should Consider Joint Supplements?

Supplements are not a replacement for prescribed medication — they are best thought of as a complementary strategy. They may be particularly useful if your arthritis is mild to moderate and not yet requiring prescription drugs; you want to reduce reliance on over-the-counter pain relief such as ibuprofen; or your diet is limited and you may not be getting sufficient anti-inflammatory nutrients from food alone.

Always consult your GP or rheumatologist before starting a new supplement, particularly if you take prescription medications, as interactions are possible.



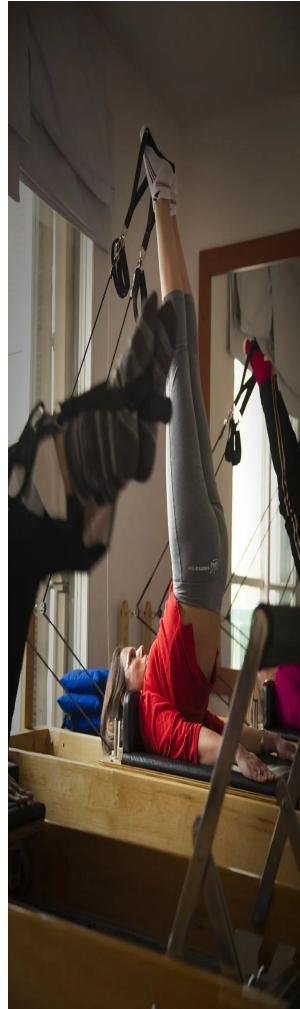
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1. Glucosamine and Chondroitin

Glucosamine and chondroitin are the most studied and most widely purchased joint supplements in the world. Both are naturally occurring compounds found in cartilage. The theory is straightforward: supplementing them may slow cartilage breakdown and reduce pain in osteoarthritis.

The evidence is mixed but leaning positive. The large GAIT trial found that glucosamine sulphate combined with chondroitin sulphate significantly reduced pain in patients with moderate-to-severe knee OA. A 2018 European meta-analysis found that prescription-grade glucosamine sulphate (crystalline glucosamine sulphate, or CGS) slowed joint space narrowing over three years — a structural benefit, not just pain relief.

****Dosage guide**** - Glucosamine sulphate: 1,500 mg per day (single or split dose). - Chondroitin sulphate: 800–1,200 mg per day. - Allow 8–12 weeks for noticeable effect — these are not fast-acting supplements. - Note: most glucosamine is derived from shellfish. Vegan alternatives (from corn fermentation) are available.



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2. Omega-3 Fish Oil

Omega-3 fatty acids — specifically EPA (eicosapentaenoic acid) and DHA (docosahexaenoic acid) — have the most robust anti-inflammatory evidence of any supplement on this list. They work by competing with omega-6 fats in the inflammatory pathway, reducing the production of pro-inflammatory leukotrienes and prostaglandins.

Multiple randomised controlled trials in rheumatoid arthritis show that fish oil supplementation significantly reduces joint tenderness, morning stiffness, and the need for NSAIDs. A 2020

Cochrane review concluded that omega-3 supplementation results in a small but clinically meaningful reduction in pain for people with RA. Benefits for OA are less pronounced but still noted.

Look for a product providing at least 2,000–3,000 mg of combined EPA + DHA per day for anti-inflammatory effects. Third-party tested products bearing the IFOS (International Fish Oil Standards) seal offer quality assurance. Algae-based omega-3 is a vegan alternative with comparable EPA/DHA content.

3. Turmeric / Curcumin

Turmeric has been used in Ayurvedic medicine for centuries, but it is only in the last two decades that clinical research has begun to validate its anti-inflammatory mechanisms. The active compound, curcumin, inhibits NF-kB — a key 'master switch' of the inflammatory response — as well as COX-2 enzymes, the same enzymes targeted by ibuprofen.

A 2021 meta-analysis of 16 randomised controlled trials found curcumin supplementation significantly reduced pain and inflammation markers compared to placebo in OA patients. The challenge is bioavailability: standard curcumin is poorly absorbed. Look for products with piperine (black pepper extract), phospholipid complex (Meriva), or nanoparticle formulations, all of which significantly improve absorption. A typical effective dose is 500–1,000 mg of bioavailable curcumin daily.

4. Vitamin D

Vitamin D deficiency is extremely common in the UK — approximately one in five UK adults has low vitamin D levels, largely due to limited sunlight during autumn and winter. Low vitamin D is associated with increased arthritis pain, worse OA outcomes, and higher RA disease activity.

UK guidance recommends everyone in the UK considers a daily 10 microgram (400 IU) supplement during autumn and winter. People with arthritis, particularly those who are housebound or have darker skin, may benefit from higher doses — 25–50 micrograms (1,000–2,000 IU) — under GP guidance. Ask your GP to test your 25(OH)D blood level as a starting point.

5. Collagen Supplements

Collagen is the main structural protein in cartilage, tendons, and ligaments. Hydrolysed collagen (collagen peptides) and undenatured type II collagen (UC-II) are the two main forms used in supplements. There is growing evidence — particularly from industry-funded trials, which should be interpreted cautiously — that collagen peptides may support cartilage repair and reduce joint pain when taken consistently.

A 2019 randomised trial published in **Nutrients** found that 40 mg of UC-II daily outperformed glucosamine and chondroitin combined for knee OA pain over 180 days. While more independent research is needed, collagen supplementation appears safe and may be a worthwhile addition, particularly for those with knee or hip OA.

6. Boswellia (Indian Frankincense)

Boswellia serrata extract is derived from the resin of the Indian frankincense tree and has impressive evidence for both OA and inflammatory arthritis. It inhibits 5-lipoxygenase, an enzyme involved in the production of leukotrienes — inflammatory molecules not targeted by standard NSAIDs.

A systematic review in the **BMC Complementary Medicine** journal found significant improvements in pain and function with *Boswellia* supplementation in knee OA. Look for extracts standardised to at least 30% AKBA (acetyl-11-keto-beta-boswellic acid), the most pharmacologically active component. Typical dose: 100–250 mg of standardised extract, twice daily.

Quick Answer

The best-evidenced supplements for joint pain in the UK are: omega-3 fish oil (2,000–3,000 mg EPA+DHA daily), glucosamine sulphate (1,500 mg/day), bioavailable curcumin (500–1,000 mg/day), vitamin D (especially if deficient), *Boswellia serrata* extract, and hydrolysed collagen peptides. Allow 8–12 weeks of consistent use to assess effects.

What to Look For When Buying Supplements

The UK supplement market is largely self-regulated. When shopping for joint supplements, prioritise: third-party testing (look for Informed Sport, NSF, or IFOS seals); standardised extracts with confirmed active compound percentages; dosages matching those used in clinical trials; and transparent labelling without proprietary blends that hide individual

ingredient amounts.

Conclusion

Food supplements for joint pain can play a meaningful supportive role in managing arthritis, particularly omega-3 fish oil, curcumin, glucosamine sulphate, and vitamin D. However, they work best as part of a broader strategy that includes an anti-inflammatory diet, appropriate exercise, prescribed medication, and regular monitoring by your healthcare team. Always discuss new supplements with your GP, especially if you are taking blood thinners, immunosuppressants, or other arthritis medications.

Q&A

Question: What is the key thing to know about food supplements for joint pain and arthritis: what really works?

Short answer: Joint supplements can complement arthritis treatment, especially for mild to moderate cases or to reduce reliance on pain relief. Glucosamine and chondroitin, omega-3 fish oil, turmeric (curcumin), vitamin D, collagen supplements, and Boswellia all have scientific evidence supporting their use for joint pain and inflammation. Always consult a GP or rheumatologist before starting any new supplement, particularly if you are on prescribed medications.

Question: When should I seek professional advice?

Short answer: If symptoms persist or worsen, pain is severe and unresponsive to usual relief, or you develop new symptoms, contact your GP, physiotherapist or rheumatology team promptly.

Question: Where can I get free arthritis support in the UK?

Short answer: Arthritis UK: 0800 5200 520. NRAS: 0800 298 7650. Free tools and guides at livingwitharthritis.org.uk.

General information only — not medical advice. Consult your GP or specialist before changing medication or starting a new programme. Emergency: 999. NHS advice: 111. Living With Arthritis · livingwitharthritis.org.uk